

Algorithmic Control and Transparency in Recommender Systems

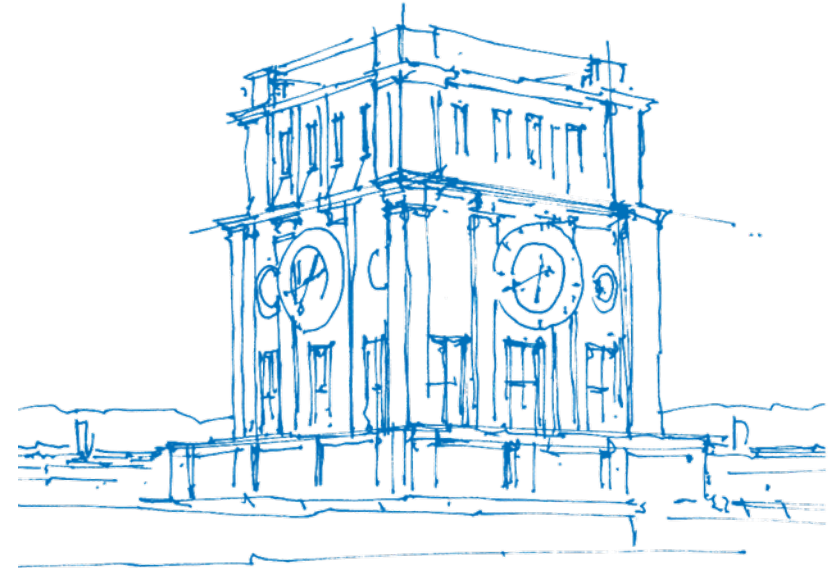
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Munich, July 6, 2026



TUM Uhrenturm

Introduction & Motivation

- **The Recommender "Black Box":**
 - Most modern recommender systems (RS) hide their inner workings (e.g. neural network rankings).
 - Causes user distrust, especially in high-stake domains like tourism (travel planning, hotel bookings).
- **Regulatory Requirements: EU Digital Services Act (DSA) [1]:**
 - **Article 27:** Mandates that platforms explain the main parameters of their RS in plain, understandable language.
 - **Article 38:** Mandates that Very Large Online Platforms (VLOPs) offer at least one profiling-free recommendation alternative.
- **Project Goal:** Design and build an interactive, transparent, and DSA-compliant tourism recommender system prototype.

Background: Recommender Systems in Tourism

- **Unique Domain Characteristics [2, 3]:**

- High financial risk and emotional investment.
- Infrequent purchasing decisions (trips are planned only once or twice a year).
- Multi-attribute search spaces (price, location, stars, transport, amenities).

- **Decision Complexity:**

- Unlike low-cost retail products (e.g. movies, books), users do not buy travel packages on impulse.
- Users require active comparison tools and want to express precise trade-offs.
- Trust and perceived control over the algorithm are essential for user acceptance.

Traditional Recommender System (RS) Strategies

- **Collaborative Filtering (CF):**
 - Predicts preferences based on ratings from similar users.
 - *Limitations in Tourism:* Cold start problems (new hotels/infrequent travelers) and lacks explanatory power.
- **Content-Based Filtering (CB):**
 - Recommends items matching a user profile built from historical behavior.
 - *Limitations in Tourism:* Poor adaptability to sudden context changes (e.g. business trip vs. family vacation).
- **Hybrid Systems [4, 5]:**
 - Combine CF and CB. However, they remain mathematical black boxes to the end-user.
 - Hard to explain, violating transparency guidelines.

HCI: The Importance of User Control & Trust

- **Being Accurate is Not Enough [6]:**

- Traditional metrics focus solely on algorithmic accuracy (e.g., RMSE).
- Ignores user factors like transparency, control, and serendipity.

- **Beyond Algorithms [7]:**

- Interactive systems should give users a direct steering wheel.
- Sliders and constraints shift the role of the user from a passive consumer to an active co-creator.

- **Explaining the User Experience [8]:**

- Combining user control with visual explanations leads to significantly higher trust, perceived recommendation quality, and system satisfaction.

UI Redesign & User Experience Challenges

Initial Usability Issues:

- Implicit strategy selection (no clear entry point).
- Cluttered landing state with premature disclosure of complex DSA transparency panels.
- Sliders scaled from 0 – 100%, confusing users into thinking weights must sum to 100%.

Applied Redesign:

- **Interactive Strategy Wizard** landing page.
- Importance scale changed to **0-10 range** to prevent percentage-summing confusion.

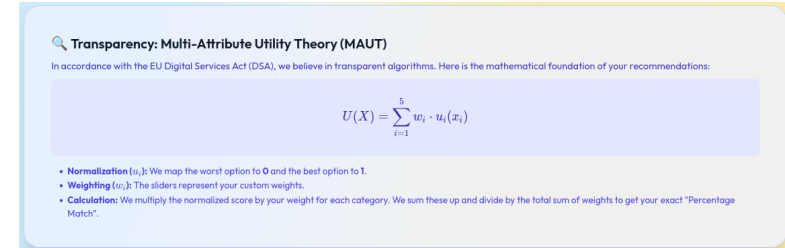


Abbildung: Previous Cluttered Transparency Board

Interactive Wizard for Algorithm Selection

- **Algorithmic State Initialization:**

- State starts as `null`.
- Recommendations, transparency boards, and controls are hidden initially to prevent cognitive overload.

- **Two Strategy Selection Cards:**

- **Weighted MAUT Card:** Focused on relative importance weighting and compensatory trade-offs.
- **Constraint-Guided Card:** Focused on hard limit rules and non-compensatory filtering.

- **Strategy Switching:**

- Simple back button ("← Switch Algorithm") to return to the wizard.
- Fast toggle tabs in the dashboard header for seamless switching.

Algorithm A: Weighted MAUT (Compensatory)

- **Compensatory Utility Scoring [9]:**

- Attributes: Price, Location, Stars, Guest Review, Transport.
- Allows high performance in one attribute (e.g. location) to compensate for low performance in another (e.g. price).

- **Weights on a 0-10 Scale:**

- Range: 0 (completely unimportant) to 10 (most important).
- Mathematically normalized by the total sum of user-specified weights:

$$U = \frac{\sum w_i \cdot u_i}{\sum w_i} \times 100\%$$

(w_i : attribute weight [0–10], u_i : normalized utility [0–1], U : match score %)

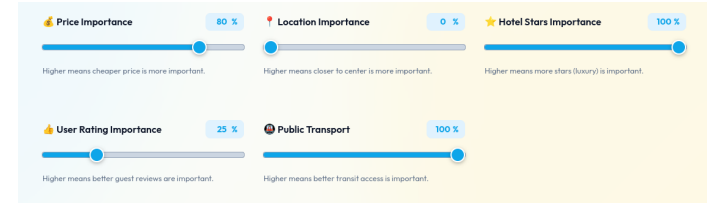


Abbildung: User Weight Sliders (0-10 scale)

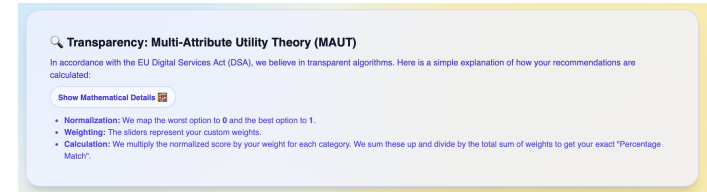


Abbildung: MAUT Transparency Explanation

Normalization Strategies in Weighted MAUT

To aggregate attributes with different units, we normalize them to a 0 – 1 scale [10]:

- **Relative Normalization (Cost attributes: Price, Distance):**

- Normalizes value v relatively based on the active city's min/max values:

$$u(v) = \frac{v_{\max} - v}{v_{\max} - v_{\min}}$$

- Normalizing cost relatively maximizes the contrast between the local hotel options.

- **Absolute Normalization (Benefit attributes: Stars, Rating, Transit):**

- Normalizes value v on fixed scale bounds to prevent artificial distortion:

$$u_{\text{stars}}(v) = \frac{v-1}{4} \quad \text{and} \quad u_{\text{rating/transport}}(v) = \frac{v-1}{9}$$

Algorithm B: Constraint-Based Filtering

- **Non-Compensatory Logic [2]:**

- User defines absolute thresholds (e.g. maximum price, minimum stars).
- A hotel must satisfy all active limits simultaneously to be recommended.
- Over-performing in one area cannot compensate for a single constraint violation.

- **DSA-Compliant Simplicity:**

- Unordered subset of matching hotels.
- Easy-to-understand inclusion/exclusion criteria.
- Complete profiling-free predictability.

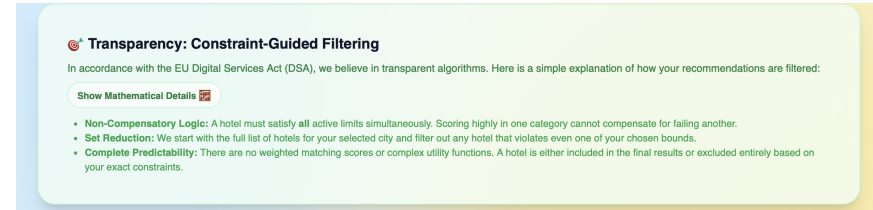


Abbildung: Constraint Filtering Interface

The Satisficing Model of Constraint Filtering

- **Logical Conjunction Formulations:**

- A hotel is recommended if and only if:

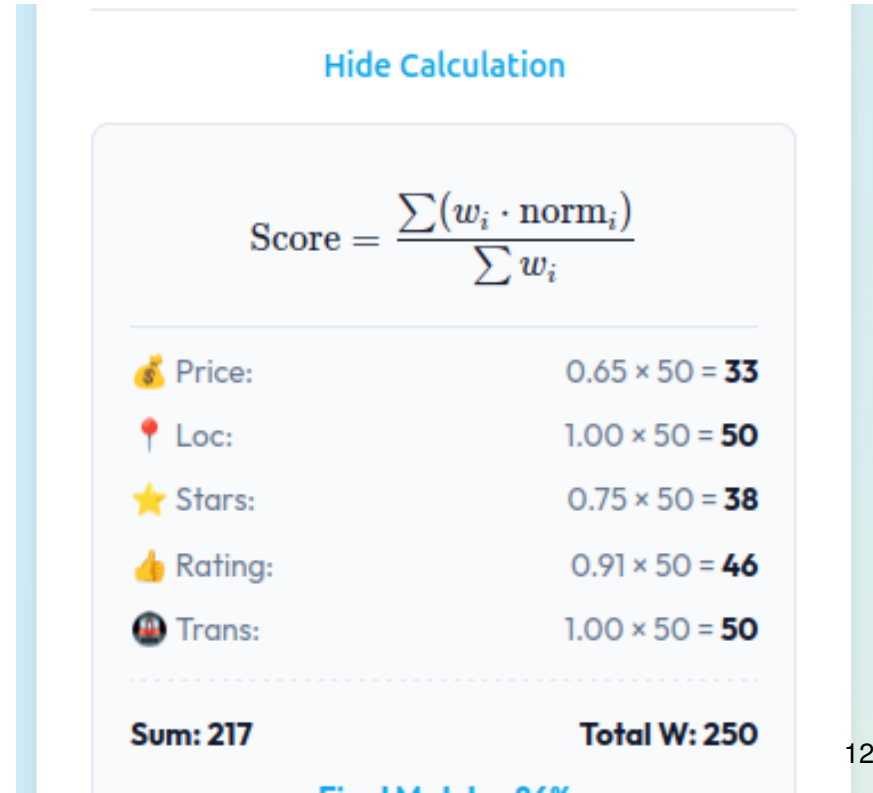
$$\text{Price} \leq P_{\max} \wedge \text{Distance} \leq D_{\max} \wedge \text{Stars} \geq S_{\min} \wedge \text{Rating} \geq R_{\min} \wedge \text{Transport} \geq T_{\min}$$

- **Psychological Background (Satisficing):**

- Based on Herbert Simon's bounded rationality theory (1957).
- Decision-makers do not seek the mathematically optimal hotel, but rather a "good enough" hotel that meets all minimal conditions.
- Drastically reduces cognitive load during selection.
- Perfect for users with rigid conditions (e.g., maximum budget).

Tiered Explainability & Transparency Panels

- **Tier 1: Dynamic Banners (Plain Text)**
 - Dynamic color schemes (Blue for MAUT, Green for Filtering).
 - Brief, plain-language summaries explaining how weights/limits are aggregated.
- **Tier 2: Algorithmic Details (TeX)**
 - Expandable KaTeX mathematical formulas for expert review and full algorithmic disclosure.
- **Tier 3: Step-by-Step Breakdown**
 - On-demand card expansions showing exact computations (Normalization $\times$ Weight).



Tiered Explanation Design Theory

- **The Explanation Dilemma [11]:**
 - Giving too little explanation makes the system look like a black box.
 - Giving too much explanation causes cognitive overload and user fatigue.
- **Tiered Disclosure (Satisfying Different User Persona):**
 - **Novice Pathway (Banners):** Brief text explanations focusing on the *meaning* of user controls. Satisfies general transparency.
 - **Expert Pathway (KaTeX Formulas):** Raw equations. Satisfies transparency audits and regulators.
 - **Curious Pathway (Math Breakdown):** Step-by-step breakdown. Explains *individual* recommendations, boosting user trust.

System Architecture & Technical Stack

- **Technology Stack:**

- React for component-based reactive state rendering.
- Vite as a fast bundler and build system.
- Vanilla CSS for a custom, glassmorphic design language.
- Client-side KaTeX rendering for mathematical formulas.

- **Real-Time Computation Flow:**

- Eliminates the latency of database queries. Real-time updates occur client-side as users drag the sliders.
- Uses React's `useMemo` hooks to cache the hotel calculations, avoiding unnecessary re-renders.
- Normalization boundaries recalculate dynamically upon city changes.

Demonstration

Real-Time Strategy Toggling & Weight Adjustments



Proposed User Study & Evaluation Design

- **Hypotheses:**

- **H1 (Perceived Control):** Users interacting with the Weighted MAUT or Constraint Filtering interfaces report a higher sense of control over recommendations compared to a baseline "black-box" interface.
- **H2 (System Trust):** Exposing the collapsible mathematical breakdown panels increases user trust in the system's recommendations without causing information overload.
- **H3 (DSA Compliance Impact):** The presence of the global explanation banner and KaTeX formulas increases user comprehension of the algorithm's inner workings.

- **Within-Subjects Laboratory Study ($N = 30$):**

- **Condition 1 (Baseline):** Traditional black-box hotel list.
- **Condition 2 (Weighted MAUT):** Slider controls with collapsible breakdowns.
- **Condition 3 (Constraint Filtering):** Threshold sliders.

Proposed User Study Evaluation Metrics

To scientifically test our hypotheses, we will gather standardized metrics:

- **System Usability Scale (SUS):**
 - 10-item Likert scale questionnaire to evaluate overall system usability.
- **ResQue (Recommender Systems Quality of Experience) [8]:**
 - Focuses on user-centric quality: perceived quality, system trust, transparency, and intent to use.
- **NASA Task Load Index (NASA-TLX):**
 - Focuses on subjective cognitive workload.
 - Measures: Mental demand, physical demand, temporal demand, performance, effort, and frustration level. Helps us verify if exposing math equations causes overload.

Summary & Conclusion

- **Successfully Completed Deliverables:**

- Developed an interactive, responsive tourism recommender SPA (React + Vite).
- Redesigned selection flow with a strategy landing wizard to prevent early transparency board clutter.
- Configured sliders on an intuitive 0-10 scale to eliminate weight-summing misunderstandings.
- Integrated tiered explainability panels (plain-language summaries, LaTeX formulas, math breakdowns).

- **Key Takeaway:**

- Demonstrates that EU DSA compliance (Article 27 & 38) can be harmonized with rich user controls, mathematical transparency, and premium UI/UX aesthetics.

Thank you for your attention!

Questions?

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